Project Design Phase-II
Data Flow Diagram & User Stories

Date	02 October 2026
Team ID	SWTID-2026-5435
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

Example: DFD Level 0 (Simplified) - Standard Laptop Ordering Flow

1.The requester opens the standard laptop catalog item on the service portal and submits the employee and laptop details.

2.Flow Designer records the request in the Laptop Requests store and routes it to the manager / budget approver.

3.The approver approves or rejects the request; the decision is returned to the flow and stored with the request.

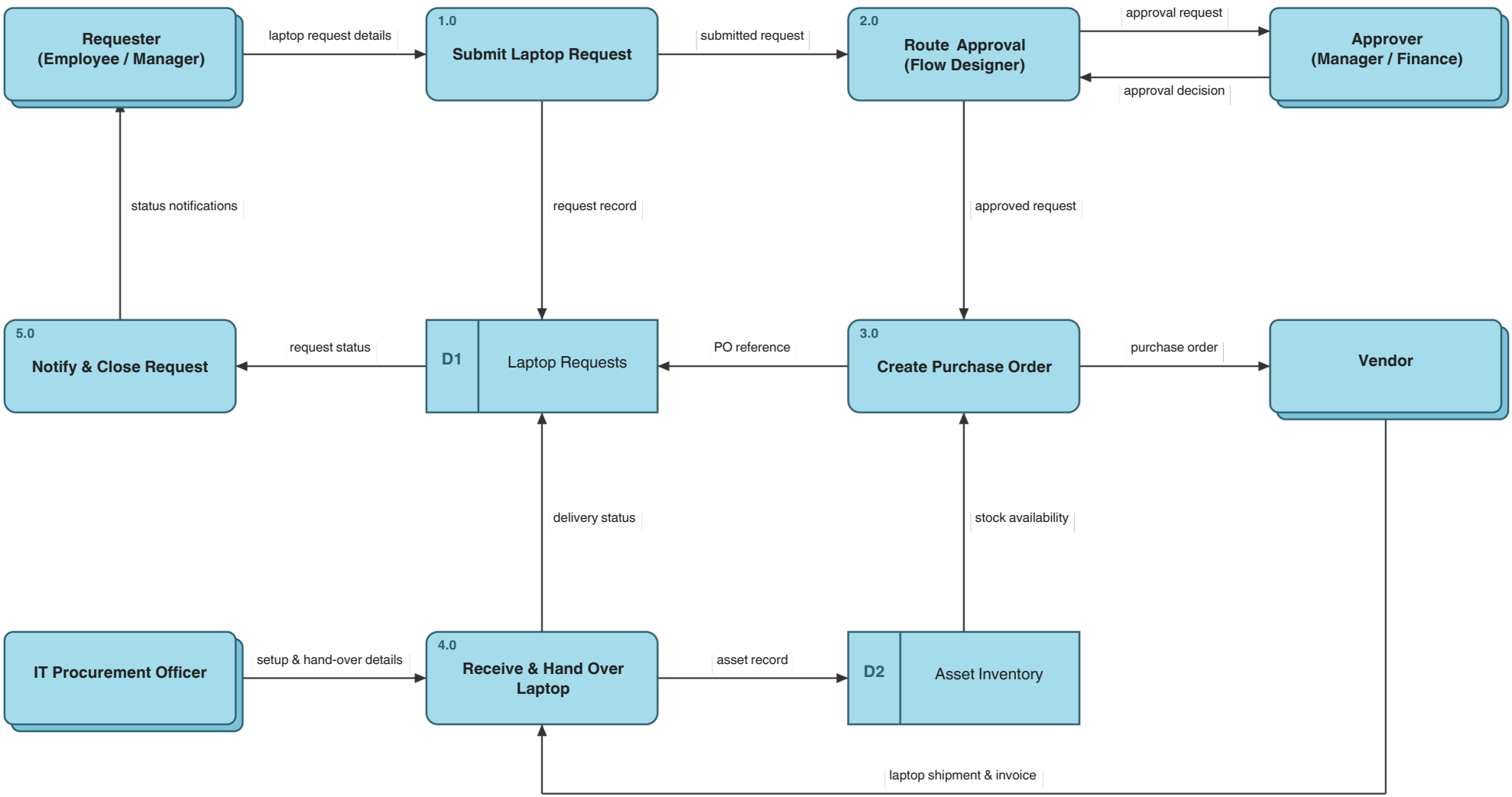
4.For approved requests, stock is checked in the Asset Inventory and a purchase order is created automatically.

5.The purchase order is sent to the vendor, who ships the laptop together with an invoice.

6. IT receives, configures and hands over the laptop, and updates the asset record and request status.

7. The requester is notified of each status change and the request is closed after hand-over.

DFD Level 0: Automated Standard Laptop Order Process



Legend: shadowed boxes = external entities; rounded boxes with numbers = processes; boxes with D1 / D2 = data stores; arrows = data flows.

User Stories

User stories for the automated standard laptop ordering process.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Requester (Employee / Hiring Manager)	Laptop Request	USN-1	As a requester, I can open the standard laptop catalog item on the service portal and submit a request by entering the employee name, role, location and joining date.	A request is created with a unique request number	High	Sprint-1
		USN-2	As a requester, I can choose the laptop model only from the approved standard models list.	Only standard models appear in the selection list	High	Sprint-1
		USN-3	As a requester, I will receive a confirmation email once I submit the request.	I receive an email with the request number and a link to the request	High	Sprint-1
	Request Tracking	USN-4	As a requester, I can view the current status of my request (pending approval, ordered, delivered, closed).	The status updates on the request page as each stage completes	Medium	Sprint-2
		USN-5	As a requester, I will be notified when my laptop is ordered, delivered and when the request is closed.	I receive a notification at every status change	Medium	Sprint-2
Approver (Manager / Finance)	Approval	USN-6	As an approver, I will receive an approval notification containing the request details.	The notification shows a request summary with Approve and Reject options	High	Sprint-1
		USN-7	As an approver, I can approve or reject a request and add comments.	My decision is saved and the flow continues or stops accordingly	High	Sprint-1
		USN-8	As an approver, I will receive a reminder if I have not responded within the set time.	A reminder is sent after the set period and escalated if still pending	Medium	Sprint-2
IT Procurement Officer	Order Processing	USN-9	As a procurement officer, I want a purchase order to be created automatically once a standard laptop request is approved.	A purchase order is generated from the request details without manual entry	High	Sprint-2
		USN-10	As a procurement officer, I can check available laptop stock before an order is placed.	The request is fulfilled from stock when available; otherwise a purchase order is raised	Medium	Sprint-2
	Delivery & Asset Management	USN-11	As a procurement officer, I can mark a laptop as received and create its asset record.	The asset record is linked to the employee and the request status is updated	High	Sprint-2
		USN-12	As a procurement officer, I can close the request after the laptop is handed over.	The request is closed and the requester is notified	Medium	Sprint-3
Administrator	Workflow Configuration	USN-13	As an administrator, I can configure approvers, approval rules and budget limits in Flow Designer without writing code.	Changes to the rules apply to new requests	High	Sprint-1
		USN-14	As an administrator, I can see failed or stalled flow executions.	Failures are logged and flagged for follow-up	Medium	Sprint-3
	Reporting	USN-15	As an administrator, I can view reports on approval time, order volume and delivery time.	A dashboard shows the metrics by period	Low	Sprint-3